

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET G — FINAL MASTERY & GRAND CHALLENGE TEST (60 MINS)****CHAPTER 01: UNDERSTANDING SOCIAL SCIENCE**

TIME ALLOWED: 60 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-G-01

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET G - Final Mastery & Grand Challenge Test):

1. The question paper comprises **25 Olympiad / NTSE / Advanced Analytical Multiple Choice Questions** of **1 mark each**.
2. Designed for **Final Curricular Mastery & In-Depth Analytical Rigour**: Target time is **60 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **GRAND CHALLENGE SPECIAL:** This final assessment tests your multi-disciplinary analytical reasoning across historical source criticism, geomorphological dynamics, constitutional jurisprudence, and microeconomic market equilibria!

Q1. An organic charcoal sample from an archaeological pit contains 25% of the initial Carbon-14 (^{14}C) activity of modern organic matter. Given that the half-life of ^{14}C is 5,730 years, what is the approximate calendar age of the sample?

- ☐ A 2,865 years ☐ B 5,730 years ☐ C 11,460 years ☐ D 17,190 years

Scholarly Concept & Analysis: After 1 half-life (5,730 yr), 50% remains; after 2 half-lives (11,460 yr), 25% remains ($N/N_0 = (1/2)^2 = 0.25$). Hence, the age is approximately 11,460 years.

Q2. Why must raw radiocarbon years (BP - Before Present) be calibrated against dendrochronological tree-ring curves (such as IntCal)?

- ☐ A Because tree rings absorb zero carbon from the atmosphere ☐ B Because atmospheric production of Carbon-14 has fluctuated over millennia due to solar cycle variations and changes in Earth's geomagnetic field ☐ C Because modern computers cannot calculate half-lives accurately ☐ D Because ancient trees were made of limestone

Scholarly Concept & Analysis: Atmospheric ^{14}C concentration is not strictly constant over time due to cosmic-ray and geomagnetic variations, requiring tree-ring calibration curves.

Q3. In archaeological stratigraphy, what is a 'Harris Matrix'?

- ☐ A A mathematical formula used to weigh gold coins ☐ B A standardized schematic diagram representing the stratigraphic sequence and temporal relationships of all archaeological contexts and strata ☐ C A microscopic lens used to inspect pollen spores ☐ D A tool used to measure soil acidity in the field

Scholarly Concept & Analysis: The Harris Matrix is a universally adopted archaeological tool that models the topological and temporal relationships of stratigraphic layers.

Q4. Which chronometric absolute dating technique measures trapped electron charges accumulated in crystalline quartz or feldspar minerals since their last exposure to sunlight?

- ☐ A Optically Stimulated Luminescence (OSL) ☐ B Potassium-Argon dating ☐ C Dendrochronology ☐ D Seriation

Scholarly Concept & Analysis: Optically Stimulated Luminescence (OSL) dates when sediment grains or quartz crystals were last exposed to sunlight prior to burial.

Q5. What is the chronological evolutionary sequence of North Indian writing scripts from the 3rd century BCE to the 10th century CE?

- ☐ A Devanagari $\rightarrow$ Gupta Brahmi $\rightarrow$ Ashokan Brahmi $\rightarrow$ Siddhamatrika ☐ B Ashokan Brahmi $\rightarrow$ Kushana Brahmi $\rightarrow$ Gupta Brahmi $\rightarrow$ Siddhamatrika (Kutila) $\rightarrow$ Nagari (Devanagari) ☐ C Kharosthi $\rightarrow$ Greek $\rightarrow$ Aramaic $\rightarrow$ English ☐ D Tamil-Brahmi $\rightarrow$ Roman $\rightarrow$ Brahmi $\rightarrow$ Indus script

Scholarly Concept & Analysis: Palaeographical evolution in North India progressed from early Ashokan Brahmi through Kushana, Gupta, and Siddhamatrika to proto-Nagari and Devanagari.

Q6. Which non-destructive scientific technique allows numismatists to determine the precise elemental composition and gold purity of ancient coins without cutting or damaging the metal?

- ☐ A X-ray Fluorescence (XRF) spectrometry ☐ B Acid dissolution testing ☐ C Hammer striking hardness test ☐ D Melting in a crucible

Scholarly Concept & Analysis: X-ray Fluorescence (XRF) spectrometry analyzes secondary fluorescent X-rays to measure metallic percentages without physical destruction.

Q7. In quantitative social survey methodology, what does a 95% Confidence Interval ($p < 0.05$) indicate regarding a researched hypothesis?

- ☐ A That the research findings are 100% false ☐ B There is less than a 5% probability that the observed statistical relationship occurred purely by random chance ☐ C That exactly 5 individuals responded to the survey ☐ D That the study was conducted over five days

Scholarly Concept & Analysis: A p-value below 0.05 indicates statistical significance, meaning the probability of finding the result by random sampling error alone is less than 5%.

Q8. In Thomas Kuhn's 'The Structure of Scientific Revolutions', what characterizes a 'paradigm shift' in scientific disciplines?

- ☐ A A minor spelling correction in a textbook ☐ B A fundamental transition where anomalies accumulate, leading to the collapse of an existing theoretical framework and the emergence of a new worldview ☐ C A government decree banning scientific research ☐ D The permanent closure of all laboratories

Scholarly Concept & Analysis: Kuhn defined paradigm shifts as revolutionary transitions where existing theoretical models are replaced by new conceptual frameworks.

Q9. Stable carbon isotope analysis ($\delta^{13}\text{C}$) on prehistoric human skeletal remains allows bioarchaeologists to distinguish between:

- ☐ A The eye color and hair length of ancient people ☐ B Diets dominated by C_3 plants (wheat, barley, tubers) versus C_4 plants (millet, maize) and tropical grasses ☐ C The exact language spoken by the individual ☐ D The individual's political party affiliation

Scholarly Concept & Analysis: $\delta^{13}\text{C}$ ratios differentiate photosynthetic pathways (C_3 vs C_4), providing precise dietary reconstruction from bone collagen.

Q10. What is the primary methodological flaw of relying exclusively on elite epigraphic court panegyrics (Prashastis) to evaluate common citizens' living standards?

- ☐ A Prashastis are written in secret invisible inks ☐ B Prashastis were commissioned by monarchs to glorify royal military conquests and aristocratic donations, systematically ignoring subaltern daily life and peasant exploitation ☐ C Inscriptions dissolve when read by common citizens ☐ D Prashastis were only written on leaves that decayed

Scholarly Concept & Analysis: Court Prashastis represent official ideological panegyrics, presenting an elite-centric narrative that omits common socioeconomic realities.

Q11. In sampling design, what is 'Stratified Random Sampling'?

- ☐ A Selecting whoever walks past the street corner first ☐ B Dividing a diverse heterogeneous population into mutually exclusive homogenous strata (e.g., by caste, income, or gender) and randomly sampling within each stratum ☐ C Surveying only friends and relatives of the researcher ☐ D Selecting participants based on their height

Scholarly Concept & Analysis: Stratified random sampling ensures proportional representation of diverse sub-groups within a complex population.

Q12. Which microscopic plant micro-fossils composed of biogenic silica (SiO_2) survive in archaeological hearths and potsherds, revealing cultivated crops even where grains decayed?

- ☐ A Phytoliths ☐ B Pollen grains ☐ C Spiders ☐ D Iron filings

Scholarly Concept & Analysis: Phytoliths are durable silica bodies formed inside plant cells that survive burning and decay, preserving evidence of ancient cereal use.

Q13. What is 'confirmation bias' in social science research?

- ☐ A Confirming that a survey form has been signed ☐ B The psychological tendency of researchers to search for, favor, and recall data that confirms their preexisting beliefs while ignoring contrary evidence ☐ C A computer printing error in a research journal ☐ D Conducting research with two independent teams

Scholarly Concept & Analysis: Confirmation bias occurs when researchers selectively gather or interpret evidence that validates their prior hypotheses.

Q14. In historical linguistic reconstruction, what does the Comparative Method achieve?

- ☐ A Proves that all human languages were invented in 1900 ☐ B Compares cognate words and phonetic sound shifts across daughter languages to reconstruct ancestral proto-languages (such as Proto-Indo-European) ☐ C Forces everyone to speak only one language ☐ D Bans the translation of ancient texts

Scholarly Concept & Analysis: The comparative linguistic method traces systematic sound changes across related languages to reconstruct hypothetical ancestral tongues.

Q15. What is 'archaeological provenance' (or proveniency)?

- ☐ A The market price of an antique sold at an auction ☐ B The exact physical, horizontal three-dimensional coordinates and stratigraphic context in which an artifact was discovered in situ ☐ C The signature of the archaeologist who washed the pottery ☐ D The museum building where an object is displayed

Scholarly Concept & Analysis: Provenance is the exact in-situ 3D find-spot and stratigraphic context of an artifact, without which archaeological value is largely lost.

Q16. Why is oral history critically triangulated with material archaeology by modern subaltern historians?

- ☐ A To prove that oral traditions are 100% fictional fairy tales ☐ B Because marginalized communities, women, and indigenous tribes often lacked access to formal writing, preserving their historical memory through oral tradition ☐ C Because printed books are illegal in tribal areas ☐ D Because archaeology has no tools to study pottery

Scholarly Concept & Analysis: Oral traditions preserve the voices and experiences of illiterate and marginalized groups excluded from official state archives.

Q17. In demographic sociological research, what is the 'Dependency Ratio'?

- ☐ A The ratio of computers to people in an office ☐ B The mathematical ratio of the dependent population (aged 0–14 and 65+) to the productive working-age population (aged 15–64) ☐ C The number of cars per kilometer of highway ☐ D The ratio of boys to girls in primary schools

Scholarly Concept & Analysis: The dependency ratio measures the demographic burden: dependent young and elderly persons relative to the working-age cohort.

Q18. What analytical distinction exists between 'material culture' and 'non-material culture' in anthropology?

- ☐ A Material culture includes physical tangible artifacts (tools, pottery, monuments), while non-material culture comprises intangible beliefs, values, norms, and languages ☐ B Material culture is only found on Mars ☐ C Non-material culture can be weighed on a physical balance ☐ D There is no conceptual difference

Scholarly Concept & Analysis: Material culture refers to physical objects and built environments; non-material culture refers to ideas, customs, and cognitive systems.

Q19. Which major scientific limitation affects Radiocarbon (^{14}C) dating beyond approximately 50,000 years Before Present?

- ☐ A Carbon atoms explode after 50,000 years ☐ B The remaining ^{14}C activity drops below 0.2% of modern activity, making it indistinguishable from laboratory background radiation ☐ C All organic materials turn into pure gold after 50,000 years ☐ D The half-life suddenly changes to zero

Scholarly Concept & Analysis: Beyond ~ 10 half-lives (50,000 yr), the residual ^{14}C signal is so minuscule that minute modern contamination produces massive dating errors.

Q20. What does a Gini coefficient of 0.0 represent in economic sociology versus a Gini coefficient of 1.0?

- ☐ A 0.0 represents absolute inequality; 1.0 represents perfect equality ☐ B 0.0 represents perfect mathematical income equality (everyone earns identical income); 1.0 represents maximum inequality (one person owns all wealth) ☐ C Both numbers represent zero income in the economy ☐ D 0.0 indicates a hyper-inflationary economy

Scholarly Concept & Analysis: The Gini coefficient ranges from 0 (perfect equality where Lorenz curve is the 45° line) to 1 (total concentration of income in one hand).

Q21. The archaeological discovery of Roman coin hoards at Arikamedu (near Puducherry) and Karur (Tamil Nadu) demonstrates:

- ☐ A That Roman legionary armies militarily conquered and colonized South India ☐ B Active maritime Indo-Roman commerce where South Indian luxury pepper and textiles were exchanged for high-value Roman gold and silver denarii ☐ C That ancient Indians could not manufacture their own bronze coins ☐ D That Roman coins were dropped accidentally by modern tourists

Scholarly Concept & Analysis: Hoards of imperial Roman coins testify to extensive Indo-Roman trade along the monsoon maritime corridor in the early centuries CE.

Q22. What is 'seriation' in relative archaeological chronology?

- ☐ A Sorting artifacts alphabetically by modern English names ☐ B Arranging archaeological artifact assemblages (such as changing pottery decorative styles) in a chronological sequence based on changes in their popular frequency ☐ C Melting pottery shards into glass beads ☐ D Throwing away all broken pots

Scholarly Concept & Analysis: Seriation orders artifacts chronologically based on stylistic evolution and fluctuating popularity curves over time.

Q23. Why is the interdisciplinary linkage between Geography and Economics critical in regional planning?

- ☐ A It proves that all geography is identical across the planet ☐ B Spatial distribution of natural resources, transport networks, and topography dictates regional trade corridors, industrial location, and comparative advantage ☐ C Both subjects use identical historical narrative poems ☐ D It eliminates the need for mathematical calculations

Scholarly Concept & Analysis: Economic geography analyzes how physical space, transit costs, and resource locations determine industrial clustering and regional prosperity.

Q24. What historiographical term describes the deliberate critical analysis of an ancient text's authorship, political context, intended audience, and potential bias?

- ☐ A Source criticism (hermeneutics) ☐ B Blind faith acceptance ☐ C Stratigraphic erosion ☐ D Numismatic casting

Scholarly Concept & Analysis: Source criticism (internal and external) critically evaluates the authenticity, intent, context, and bias of historical documents.

Q25. In social sciences, an ethical commitment to democratic pluralism requires researchers to:

- ☐ A Falsify data to favor the ruling government ☐ B Respect and document diverse perspectives, protect participant confidentiality, and uphold evidence-based integrity regardless of political pressure ☐ C Refuse to publish any findings that criticize societal inequalities ☐ D Exclude all women and marginalized groups from research samples

Scholarly Concept & Analysis: Ethical social research requires intellectual honesty, protection of human subjects, and rigorous representation of diverse viewpoints.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET G)

01 A B C D	02 A B C D	03 A B C D	04 A B C D	05 A B C D
06 A B C D	07 A B C D	08 A B C D	09 A B C D	10 A B C D
11 A B C D	12 A B C D	13 A B C D	14 A B C D	15 A B C D
16 A B C D	17 A B C D	18 A B C D	19 A B C D	20 A B C D
21 A B C D	22 A B C D	23 A B C D	24 A B C D	25 A B C D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & IN-DEPTH SCHOLARLY EXPLANATIONS (SET G)

Q. No.	Correct	Curricular Concept & Scholarly Analysis
1	C	After 1 half-life (5,730 yr), 50% remains; after 2 half-lives (11,460 yr), 25% remains ($N/N_0 = (1/2)^2 = 0.25$). Hence, the age is approximately 11,460 years.
2	B	Atmospheric ^{14}C concentration is not strictly constant over time due to cosmic-ray and geomagnetic variations, requiring tree-ring calibration curves.
3	B	The Harris Matrix is a universally adopted archaeological tool that models the topological and temporal relationships of stratigraphic layers.
4	A	Optically Stimulated Luminescence (OSL) dates when sediment grains or quartz crystals were last exposed to sunlight prior to burial.
5	B	Palaeographical evolution in North India progressed from early Ashokan Brahmi through Kushana, Gupta, and Siddhamatrika to proto-Nagari and Devanagari.
6	A	X-ray Fluorescence (XRF) spectrometry analyzes secondary fluorescent X-rays to measure metallic percentages without physical destruction.
7	B	A p-value below 0.05 indicates statistical significance, meaning the probability of finding the result by random sampling error alone is less than 5%.
8	B	Kuhn defined paradigm shifts as revolutionary transitions where existing theoretical models are replaced by new conceptual frameworks.
9	B	$\delta^{13}\text{C}$ ratios differentiate photosynthetic pathways (C_3 vs C_4), providing precise dietary reconstruction from bone collagen.
10	B	Court Prashastis represent official ideological panegyrics, presenting an elite-centric narrative that omits common socioeconomic realities.
11	B	Stratified random sampling ensures proportional representation of diverse sub-groups within a complex population.
12	A	Phytoliths are durable silica bodies formed inside plant cells that survive burning and decay, preserving evidence of ancient cereal use.
13	B	Confirmation bias occurs when researchers selectively gather or interpret evidence that validates their prior hypotheses.
14	B	The comparative linguistic method traces systematic sound changes across related languages to reconstruct hypothetical ancestral tongues.
15	B	Provenance is the exact in-situ 3D find-spot and stratigraphic context of an artifact, without which archaeological value is largely lost.
16	B	Oral traditions preserve the voices and experiences of illiterate and marginalized groups excluded from official state archives.
17	B	The dependency ratio measures the demographic burden: dependent young and elderly persons relative to the working-age cohort.
18	A	Material culture refers to physical objects and built environments; non-material culture refers to ideas, customs, and cognitive systems.
19	B	Beyond ~10 half-lives (50,000 yr), the residual ^{14}C signal is so minuscule that minute modern contamination produces massive dating errors.

Q. No.	Correct	Curricular Concept & Scholarly Analysis
20	B	The Gini coefficient ranges from 0 (perfect equality where Lorenz curve is the 45° line) to 1 (total concentration of income in one hand).
21	B	Hoards of imperial Roman coins testify to extensive Indo-Roman trade along the monsoon maritime corridor in the early centuries CE.
22	B	Seriation orders artifacts chronologically based on stylistic evolution and fluctuating popularity curves over time.
23	B	Economic geography analyzes how physical space, transit costs, and resource locations determine industrial clustering and regional prosperity.
24	A	Source criticism (internal and external) critically evaluates the authenticity, intent, context, and bias of historical documents.
25	B	Ethical social research requires intellectual honesty, protection of human subjects, and rigorous representation of diverse viewpoints.